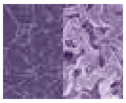
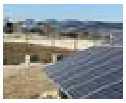


New alternatives to lithium
Sodium-based material yields stable alternative to lithium-ion batteries.
https://dgit.in/dec21-39

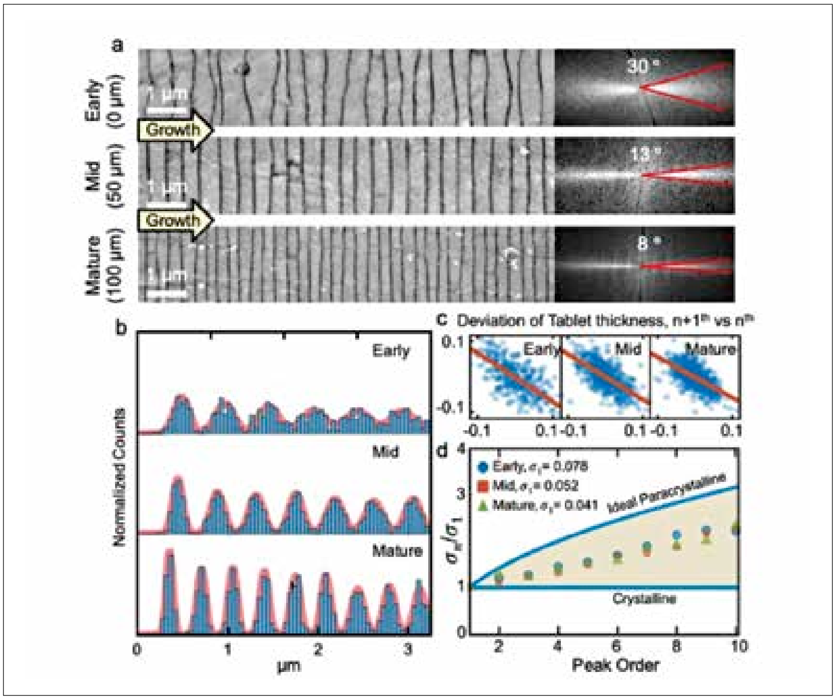


Brownfields to Greenfields
Solar farms built on top of defunct landfills are transforming dry land into lush green fields.
https://dgit.in/dec21-40

thickness of the layers is regulated. This means that, for example, if a layer of nacre is thick, the layers that are formed after that are much finer to compensate for the anomaly that happened during the formation of that thick layer. As reported in the research paper, this process is highly regulated. When looked at under an electron microscope, the seemingly random process of all the layers stacking up displayed a pattern that was later termed as 'pink noise' or 1/f behaviour. What this means is that this seemingly random process of stacking layers of nacre is rather a mathematical progression, which leads to regulation of layers, in this case, 20 at a time. This may seem to be too little but considering the number of layers of nacre that are stacked to make a decent-sized pearl, the number goes into the thousands, and this corrective action is enough to maintain symmetry.

The way in which the scientists discovered and studied this process was that they took specimens of nacreous non-bead-cultured Akoya "keshi" pearls, which were made by molluscs in the Broken Bay Pearl Farm, off the eastern coast of Australia. They then used a diamond wire saw to cut out the subsection of the pearl, which were 3 to 5 millimetres in diameter. The inspection was carried out under an electron microscope, using the Raman spectroscopy method.

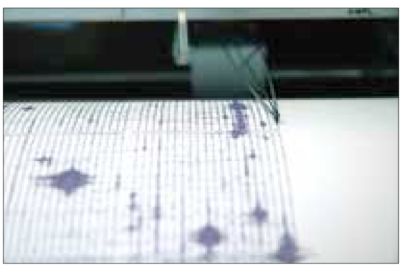
One of the pearls under study was found to have developed precisely 2,615 layers of nacre over a period of 548 days. In the pearl under study, the nacre was responsible for making 87 per cent of the pearl's material. Further observations revealed the 'pink noise' pattern which we mentioned earlier. The researchers, commenting on the nature of the corrective and trigger behind the corrective action, mentioned in the paper that, "Mechanistically, the source of this correlative correction is not clear—it may be explained by a partial viscosity of the $CaCO_3$ or the precursor phase upon deposition under hydrostatic pressure."



Source: Proceedings of National Academy

Nacre's structural variation

It was concluded that this corrective action taken by the mollusc to maintain the shape of the pearl was beneficial to its overall shape and the subsequent aesthetic appeal gave rise to another problem, which was of variation. This corrective action inside the mollusc created a cascade of stacking correction, which in turn led to each pearl having its own variation, increasing the level of uniqueness that each pearl exhibits. During the entire process, as elucidated elaborately in the paper, the "molluscs strike a balance between preserving translational symmetry and minimising thickness variation of layers by creating a paracrystal with medium-range order".



'Pink Noise' is also seen in earthquakes

WHY IS IT SIGNIFICANT?

This discovery by this team of researchers is significant for scientists and their fellow researchers worldwide. It could open up a whole new avenue for the research and developments to be made in the study of next-generation, high-performance nanomaterials. The main reason behind this is the strength of nacre, which places it to be about 3,000 times stronger than the materials which constitute it, which are calcium, and protein, amongst the many others.

This, coupled with an understanding of how it is naturally arranged to make one of the most fascinating gems, could prove to be instrumental in developing materials and manufacturing techniques for solar panels and other materials, which could be used to advance space science and technology. Laura M. Otter, one of the researchers in the team, in a statement published on ScienceNews, said, "These humble creatures are making a super light and super tough material so much more easily and better than we do with all our technology," further highlighting the importance of this discovery. d